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MT249
Practical modern statistics

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 14 /April /2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed: 90 Minutes**

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part1: Q-1 [15 marks]

A- [10 marks] Answer the following 7 questions as True or False (1 mark),) and write the correct statement if the answer is false(1mark) :

1. For a moving average of order 2q + 1, the values for the first q and the last q time points can be calculated.
2. ARMA (0, 0) model is the white noise model.
3. One way to reduce the noise is to replace each value X_t with the average of X_t and its neighboring values.
4. When under-smoothing the data then there is a risk that variations of interest & important details in the trend smooth out.
5. Techniques such as differencing or normalization may be needed to induce stationarity time series.
6. The optimal choice for the smoothing parameter α is the value which minimizes the sum of squared errors(SSE).
7. The sample autocorrelation at lag k is calculated using the k pairs of points.

B- [5 marks] Fill in the blanks in the following 5 statements:

1- The time series that is not seasonal and has no systematic variation in the level is ---------- in mean.
2- The ---------- of a time series into its component parts, such as trend, seasonal, and irregular components, can help identify patterns in the data.
3- ---------- the series is an approach to remove the trend without estimating it explicitly.
4- A ---------- is a regular pattern that repeats at fixed intervals.
5- ---------- **is** variation in a time series within one year that is repeated more or less regularly.

Part2: Each question is worth 15 marks. Answer the following questions:

Q–1: [4+4+3+4 marks]:

For the time plot for the time series of weekly reports of rotavirus infection in Figure 1.

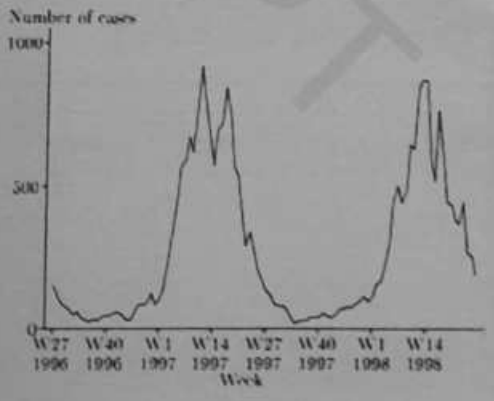


Figure 1

(a) Comment briefly on two of the patterns you notice in this time series.

Two transformations of the time plot of monthly sales by a company between January 1965 and May 1971 are shown in Figure 2. Figure 2(a) shows the time plot of the logarithms of the monthly sales. The time plot of the square roots of the monthly sales is shown in Figure 2(b).

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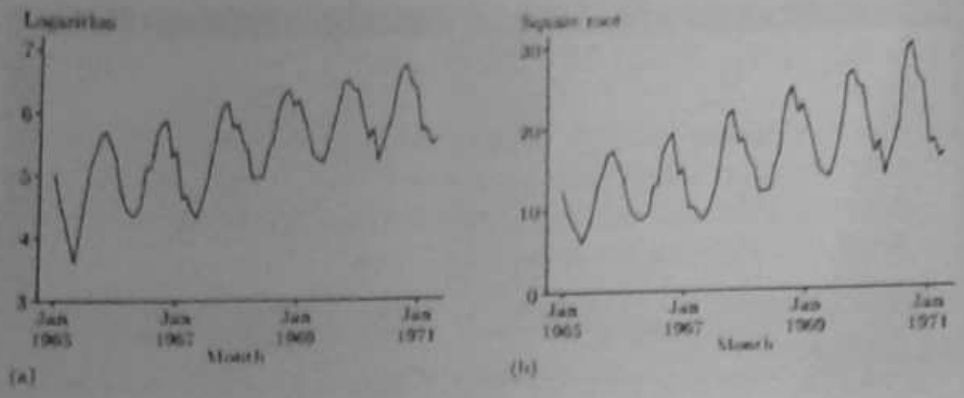


Figure 2

(b) In your opinion, which of the two-time series in Figure 2 would it be more appropriate to describe by an additive model?

(c) In any time series, the autocorrelation at lag 0 is equal to 1. Explain why this is so.

(d) Give an example of a sequence of numbers that represents the first few values of a seasonal component for a quarterly additive decomposition time series model.

Q–2: [3+3+3+6 marks]:

(a) How the order of the moving average should be chosen when smoothing a time series?

(b) Table 1 shows the first three values of a time series. Use simple exponential smoothing with $\alpha = 0.75$ and starting value $\hat{x}_1 = x_1$ to obtain the1-step ahead forecast of x_4.

t	x_t
1	0.52
2	0.36
3	0.48

(c) For a given set of starting values, the optimal values for the smoothing parameters were found to be $\alpha = 0.81$, $\gamma = 0.09$. Briefly interpret these values in terms of dependence on observations in the recent and distant past.

(d) Figure 3(a) shows the time plot of monthly average temperatures in Central England. Figure 3(b) shows the 1-step ahead forecasts for the year up to January 2005 obtained by using the simple, Holt and Holt–Winters exponential smoothing methods.

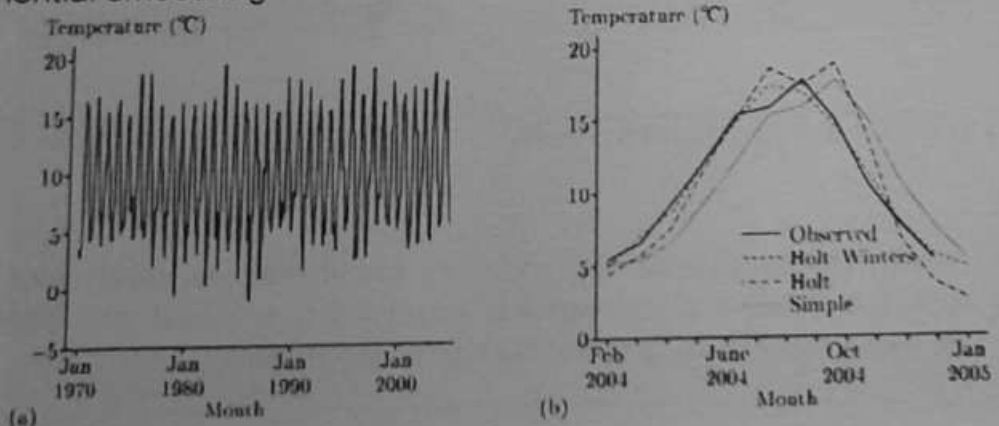


Figure 3

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1- The SSEs for the three methods are 3387, 799 and 3300 (not necessarily in that order). Which SSE corresponds to the Holt–Winters model, and why?
2- The forecasted monthly temperatures for February 2005 were 5.3◦C using simple exponential smoothing, 4.6◦C using Holt's method, and 4.8◦C using the Holt–Winters method. Which forecast is likely to be the most reliable? Briefly justify your answer.

Q–3: [5x3 marks]:
Temperatures of a chemical process (in ◦C) were measured at hourly intervals. Figure 4(a) shows the time plot of the temperatures, and Figures 4(b) and 4(c) show time plots of the first and second differences, respectively.

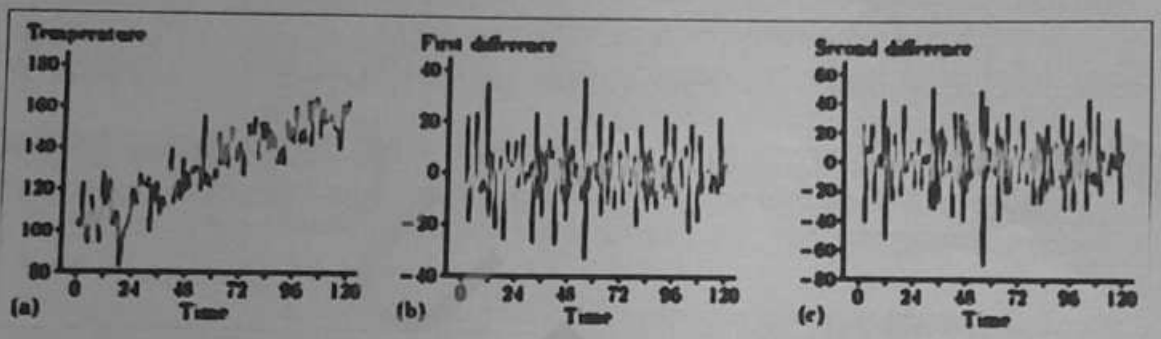


Figure 4

(a) Identify the order of differencing that appears to be required to obtain stationarity in mean. Justify your answer.
(b) Does the differenced series that you identified in part (a) appear to be stationary in variance? Justify your answer.

The correlogram Figure 5(a) and the partial correlogram Figure 5(b) for a stationary time series are shown below in Figure 5

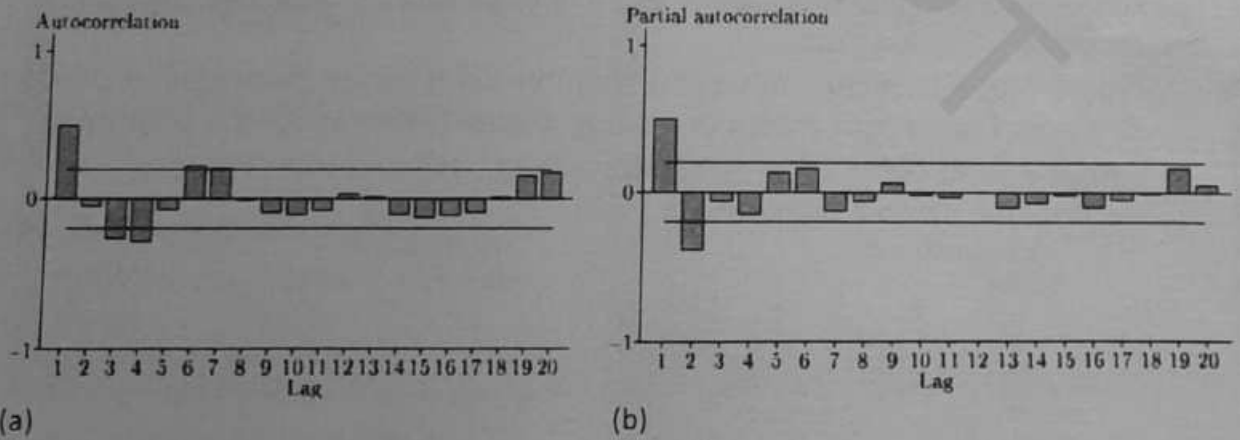


Figure 5

(c) Identify two plausible ARMA models for this time series. Justify your choices.
(d) Apply the principle of parsimony to select one or two 'best' candidate models.
(e) List these models using the ARIMA(p, d, q) notation.

End of questions

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